

1) Suppose:

$$\text{Var}(x) = 25$$

$$\text{Var}(y) = 9$$

$$\text{Cov}(x, y) = 12$$

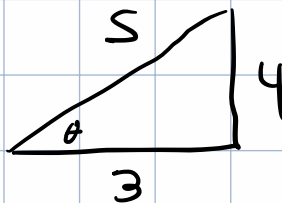
$$\langle x, y \rangle = E[\tilde{x}\tilde{y}]$$

Compute:

- $\|x\| = \sqrt{25} = 5$
- $\|y\| = \sqrt{9} = 3$
- Correlation = $\cos \theta(x, y)$

$$= \frac{\text{Cov}(x, y)}{(\sqrt{\text{Var}(x)})(\sqrt{\text{Var}(y)})}$$

$$= \frac{12}{5 \cdot 3} = \frac{4}{5}$$



- $\theta = \cos^{-1}\left(\frac{4}{5}\right) \approx .644$

2) Construct the covariance matrix given

$$\text{Var}(x) = 4$$

$$\text{Var}(y) = 16$$

$$\text{Cov}(x, y) = -6 = \text{Cov}(y, x)$$

$$\Sigma = \begin{bmatrix} \text{Var}(X) & \text{Cov}(Y, X) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{bmatrix} = \begin{bmatrix} 4 & -6 \\ -6 & 16 \end{bmatrix}$$

Is Σ symmetric? Why should it be?

Σ is symmetric because $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

3) Without doing any calculations, explain why PCA always produces a diagonal covariance matrix

Principal Component Analysis always produces a diagonal covariance matrix because it rotates the covariance matrix into a reference frame where basis vectors (principal components) are orthogonal. When vectors are orthogonal, their covariance is zero. Since the covariance matrix is comprised of the variance of the basis vectors on the diagonal, and the covariance of the vectors on the off-diagonals, that gives us a matrix of the form:

$$\Lambda = \begin{bmatrix} \text{Var}(x_0) & 0 & 0 & \dots & 0 \\ 0 & \text{Var}(x_1) & 0 & \dots & 0 \\ 0 & 0 & \text{Var}(x_2) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \text{Var}(x_n) \end{bmatrix}$$

Which means that Λ is diagonal in this reference frame

Please check that I'm doing PCA correctly here for the data from ①

$$\Sigma = \begin{bmatrix} 25 & 12 \\ 12 & 9 \end{bmatrix} \xrightarrow{\text{PCA}} \det \begin{bmatrix} 25-\lambda & 12 \\ 12 & 9-\lambda \end{bmatrix} = 0$$

$$\begin{aligned} \det(\Sigma) &= (25-\lambda)(9-\lambda) - 144 = 0 \\ &= \lambda^2 - 34\lambda + 225 - 144 = 0 \\ &= \lambda^2 - 34\lambda + 81 = 0 \end{aligned}$$

$$\begin{aligned} \lambda &= \frac{34 \pm \sqrt{1156 - 4 \cdot 81}}{2} \\ &= \frac{34 \pm \sqrt{1156 - 324}}{2} \end{aligned}$$

$$= \frac{34 \pm 2^3 \sqrt{13}}{2}$$

$$\lambda = 17 \pm 2^2 \sqrt{13}$$

$$\Lambda = \begin{bmatrix} 17 + 2^2 \sqrt{13} & 0 \\ 0 & 17 - 2^2 \sqrt{13} \end{bmatrix}$$

$$\begin{array}{r} 25 \\ -144 \\ \hline 81 \end{array}$$

$$\begin{array}{r} 34 \\ \times 34 \\ \hline 136 \\ 1020 \\ \hline 1156 \end{array}$$

$$\begin{array}{r} 81 \\ \times 4 \\ \hline 324 \end{array}$$

$$\begin{array}{r} 1156 \\ - 324 \\ \hline 832 \end{array}$$

$$\begin{array}{r} 832 > 2 \\ 416 > 2 \\ 208 > 2 \\ 104 > 2 \\ 52 > 2 \\ 26 > 2 \\ 13 \end{array}$$

4

$$\text{Cov}(A) = V \Lambda^2 V^T$$

$$(17+4\sqrt{13})^2$$

$$289 + 2 \cdot 17 \cdot 4 \cdot \sqrt{13} + 16 \cdot 13$$

$$\begin{array}{r} 17 \\ \times 17 \\ \hline 119 \\ 170 \\ \hline 289 \end{array}$$

$$\Sigma = \begin{bmatrix} 9 & 6 \\ 6 & 9 \end{bmatrix} \quad \text{find } \Lambda$$

$$\Sigma = V \Lambda V^T$$

$$\det \begin{bmatrix} 9-\lambda & 6 \\ 6 & 9-\lambda \end{bmatrix} = 0$$

$$\begin{array}{r} 81 \\ - 36 \\ \hline 45 \end{array}$$

$$\lambda^2 - 18\lambda + 81 - 36 = 0$$

$$\lambda^2 - 18\lambda + 45 = 0$$

$$(\lambda - 15)(\lambda - 3) = 0$$

$$\lambda = \begin{bmatrix} 15 \\ 3 \end{bmatrix} \quad \Lambda = I \lambda = \begin{bmatrix} 15 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 4 & -6 \\ -6 & 16 \end{bmatrix} \quad \text{find } \Lambda$$

$$\det \begin{bmatrix} 4-\lambda & -6 \\ -6 & 16-\lambda \end{bmatrix} = 0$$

$$(4-\lambda)(16-\lambda) - 36 = 0$$

$$\lambda^2 - 20\lambda + 64 - 36 = 0$$

$$\lambda^2 - 20\lambda + 28 = 0$$

$$\begin{array}{r} 3 \\ 28 \end{array}$$

$$\lambda = \frac{20 \pm \sqrt{400 - 4 \cdot 28}}{2}$$

$$\lambda = \frac{20 \pm \sqrt{400 - 112}}{2}$$

$$\lambda = \frac{20 \pm \sqrt{288}}{2}$$

$$\sqrt{288} \approx 17.0$$

$$\lambda \approx \begin{bmatrix} \frac{20 + 17.0}{2} \\ \frac{20 - 17.0}{2} \end{bmatrix} = \begin{bmatrix} 18.5 \\ 1.5 \end{bmatrix}$$

$$\Lambda \approx \begin{bmatrix} 18.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$

$$\begin{array}{r} \times 4 \\ \hline 112 \end{array}$$

$$\begin{array}{r} 3910 \\ 400 \\ -112 \\ \hline 288 \end{array}$$